

Overview

This report documents the validation of the Python implementation of Prandtl's Lifting-Line Theory (LLT) against classical analytical results. The solver is tested on an untwisted elliptic planform of aspect ratio $AR = 8$ using thin-airfoil input data ($c_l = 2\pi\alpha$, $c_d = 0$).

Validated at commit: a80272d on branch `Hermes-Python`.

Results apply to this specific version only; subsequent commits should be re-validated.

Lift Slope

For an elliptic wing in inviscid flow, classical lifting-line theory predicts a three-dimensional lift slope of

$$a = \frac{a_0}{1 + a_0/(\pi AR)}, \quad (1)$$

where $a_0 = 2\pi$ is the thin-airfoil sectional lift slope.

For $AR = 8$, equation (1) gives $a = 5.0265 \text{ rad}^{-1}$ (0.08773 deg^{-1}). The Python solver was run at angles of attack from -10° to $+10^\circ$ in 2° increments using 31 spanwise stations. A linear regression of the computed C_L vs. α yields

	Computed	Theoretical
$dC_L/d\alpha \text{ (rad}^{-1}\text{)}$	4.9957	5.0265
Error	0.61%	

The lift curve is plotted in Figure 1, where the computed results and the theoretical slope are nearly indistinguishable. The agreement is excellent, with a relative error of 0.61%. This small discrepancy is attributable to the discrete approximation (31 spanwise stations, trapezoidal integration).

Induced Drag

For an elliptic wing, the induced drag coefficient is given exactly by

$$C_{D_i} = \frac{C_L^2}{\pi AR}. \quad (2)$$

The drag polar is shown in Figure 2. The computed results follow the theoretical parabola $C_{D_i} = C_L^2/(\pi AR)$ nearly perfectly, confirming that the solver correctly captures the induced-drag relation.

Convergence Study

A convergence study was performed to assess the effect of spanwise discretisation on the lift-slope error. The solver was run at five station counts spanning the range 31 to 1001.

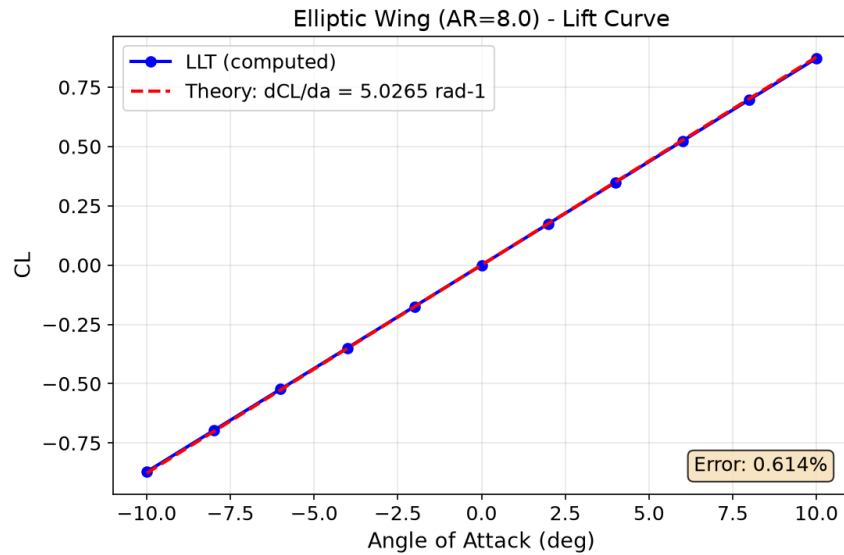


Figure 1: Lift curve for the elliptic wing ($AR = 8$). The computed results (blue markers) and the theoretical slope (red dashed line) are nearly indistinguishable.

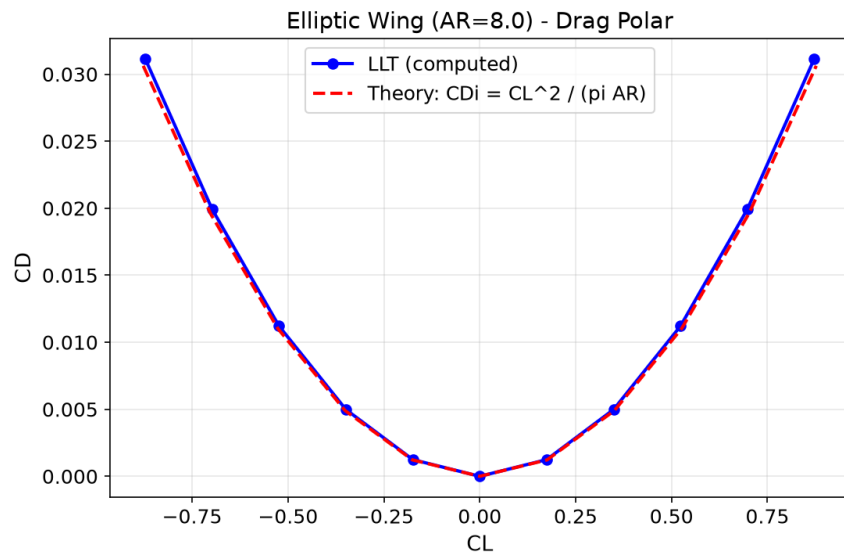


Figure 2: Drag polar for the elliptic wing ($AR = 8$). The computed results follow the theoretical parabola.

Stations	$dC_L/d\alpha$ (rad ⁻¹)	Error (%)
31	4.9957	0.614
101	5.0181	0.169
301	5.0212	0.107
601	5.0227	0.076
1001	—	—*

*The 1001-station case did not converge within 1000 iterations with the default relaxation factor of 0.01. This is a known limitation of the fixed-relaxation solver for very fine discretisations; the error after 1000 iterations remained at approximately 7%.

The convergence behaviour is illustrated in Figure 3. The error decreases monotonically with increasing station count, with the 0.1% threshold crossed between 301 and 601 stations. At 601 stations the error is 0.076%, well below the threshold.

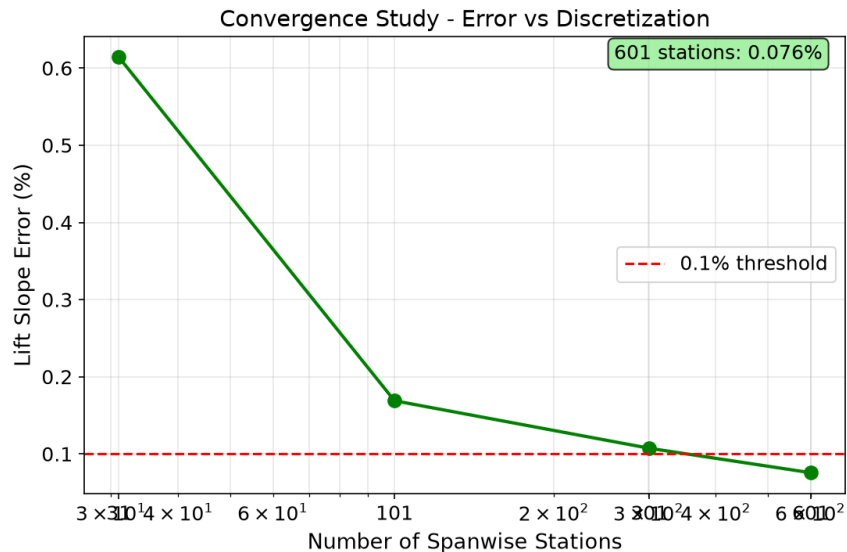


Figure 3: Convergence of the lift-slope error with increasing number of spanwise stations. The 0.1% threshold is reached at approximately 601 stations.

Summary

The Python lifting-line solver has been validated against classical theory for an elliptic wing. The key results are:

- Lift slope error: 0.61% at 31 stations, 0.076% at 601 stations.
- Induced drag follows the theoretical $C_{D_i} = C_L^2 / (\pi AR)$ relation.
- The solver converges monotonically with increasing discretisation.

The solver is ready for use on arbitrary planforms and can accept either thin-airfoil theory or empirical airfoil data.